

Metage2Metabo PostaViz: a Python package for exploring, visualising, and comparing the metabolic potential of microbial communities

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Summary

Microbial communities consist of up to thousands of distinct microbial populations, each characterized by its genomic DNA sequences, and all sharing a habitat and environmental conditions. The word *microbiome* describe the holistic concept associated all the previous components ([Berg et al., 2020](#)). Characterising the populations in samples and understanding both their roles of and interactions within the microbiome require a combination of experimentation, high-throughput data acquisition and computational models ([Klitgord & Segrè, 2011](#)). The role of microbial populations can be abstracted by the study of their *metabolism*, encompassing all biochemical reactions that cells may perform. The genome encodes the genetic information associated to the metabolism, and can therefore be queried to estimate the metabolic capabilities of microorganisms, represented as *metabolic networks*. Because such metabolism is highly redundant, depends on the environment, and because of the size of microbial communities, computational models are needed and predict, from the networks, the possible behaviours of and interactions within microbial populations in given environmental conditions. Comparing the outcomes of such predictive models in multiple samples further increases the difficulty of integrating results into actionable hypotheses. Visualisation and integration of metadata can help perform such comparison. They need to be as customisable as possible to facilitate exploration by end-users.

Statement of need

Metage2Metabo-PostaViz (M2M-PostAViz) is a Python package that performs analyses on the predictions generated by the metabolic-modelling tool [Metage2Metabo](#) (M2M) ([Belcour et al., 2020](#)). M2M screens the metabolic potential of a microbial community represented as a collection of genome-scale metabolic networks. When working with cohorts of hundreds or thousands of samples, one has to run the tool as many times as there are samples, then analyse the results of the model. The tool's outputs are, for each community, several data frames describing the role of each microorganism with respect to the whole community's functions. Properly comparing all samples requires combining all the outputs, and taking into account sample metadata describing individuals lifestyle or clinical information for instance.

M2M-PostaViz integrates all such data and provides a visualisation interface that permits exploration through custom plot generation and statistical tests. The underlying data treatment was optimised in order to deal with large numbers of samples without impeding user experience. M2M-PostaViz notably permits a pre-treatment and storage of the data such that future exploration can be launched in a computationally efficient manner. Exploration is performed at several levels: molecules (metabolites) that may or may not be producible across samples, microorganisms that may have different behaviours across samples depending on interactions

41 with other community members, or more general overviews of the community functions. The
42 tool works as a local web-based application.

43 Overall, M2M-PostaViz was designed to address three needs: (i) comparing large metagenomic
44 datasets using a metabolic modelling approach; (ii) integrating additional data into the
45 analysis, such as microbial community composition and associated metadata; and (iii) making
46 these analyses accessible to non-specialists through a graphical user interface that supports
47 customisable workflows."

48 State of the field

49 Metabolic modelling is widely used to determine the roles of microorganisms in microbial
50 communities (Cerk et al., 2024): what molecules they can produce, which interactions are
51 likely to happen depending on the environmental conditions... Many models rely on integer
52 linear programming optimisations (García-Jiménez et al., 2021) and some alternatives use
53 Boolean abstractions (Frioux et al., 2018) (Belcour et al., 2020) or probabilistic approaches
54 (Bernstein et al., 2019) to provide predictions on community behaviours. Key questions are
55 scalability to large communities, and also integration of these predictions when analysing many
56 samples or community compositions, which is more and more frequent as large metagenomic
57 cohorts get published (Asnicar et al., 2025).

58 In practice, each microorganism is abstracted by a collection of biochemical reactions it may
59 perform according to its genomic information, thus forming a network connecting transformed
60 molecules (Cerk et al., 2024). A community of microorganisms is therefore represented as
61 a collection of such networks, referred to as genome-scale metabolic networks. A metabolic
62 model will provide predictions on the possible behaviour of microorganisms and communities in
63 defined simulation conditions. Boolean abstractions, as used by M2M, ensure the scalability of
64 predictions; although the model is qualitative and does not quantify interactions, such methods
65 are considered a reasonable proxy for more quantitative models (Kruse & Ebenhöf, 2008).

66 Software design

67 Data integration and storage

68 Each sample has to be run in M2M prior analysis. Inputs to the application are a set of directories
69 generated by M2M for each sample, composed of several tabulated and json files. Additional
70 inputs to M2M-PostaViz include relative abundance of microbes in each sample, used to weigh
71 the predictions, metadata associated to the samples and possibly metadata related to molecule
72 descriptions and microbial taxonomy of the corresponding metabolic networks present in the
73 communities.

74 Reading and integrating all data is computationally demanding when considering several
75 hundreds or several thousands of samples (and M2M outputs directories). Taking into account
76 that users are likely to explore the same data across several runs of the application, efforts
77 were done to efficiently store the required and processed data such that only the first run takes
78 time and the future ones directly load pre-processed data. The parquet format is used to store
79 all this information as a database limiting the use of memory for accesses.

80 Application content

81 The application opens as a multi-tab browser page where the first one is an overview of the
82 data that summarises it and enables several first analyses customisable by variables of metadata.
83 Two tabs provide analyses centered of microbe roles and molecules respectively (see below)
84 and a last one summarises the metadata and permits customising variable types to fine-tune
85 analyses and plots.

86 Exploration of microorganism roles

87 The second tab of the application focuses on the role of microorganisms in the production
88 of metabolites across samples. The same species, and thus the same metabolic network can
89 appear in several samples, but behave differently because of interactions with other microbial
90 populations. In addition, microbial species can be grouped according to their taxonomy,
91 enabling to consider not only the role of a metabolic network but the role of all those falling
92 in taxonomic groups of different levels (phylum, family, genera...). Again, sample metadata
93 variables can refine the analysis, enabling users to compare groups or filter certain samples.

94 Exploration of metabolite production across samples

95 The third tab of the application is dedicated to the analysis the metabolites, that can also be
96 groups of families if a proper ontology is provided. Focusing on metabolites enables a targeted
97 analysis in order to compare samples or groups of samples on specific metabolic functions.

98 Research impact statement

99 M2M-PostaViz builds on the widely used M2M framework, which has been cited more than
100 one hundred times and remains actively maintained, by addressing key barriers that currently
101 limit the broader adoption of large-scale metabolic modelling. While M2M enables powerful
102 predictions of metabolic complementarity and community-level functions, its complexity and
103 the volume of generated outputs require substantial expertise, restricting its use to a limited
104 and expert. audience.

105 M2M-PostaViz overcomes these limitations by providing an interactive and user-friendly graphical
106 interface that enables systematic exploration, comparison, and interpretation of M2M outputs
107 across large metagenomic datasets consisting of up to thousands samples. The tool integrates
108 metabolic modelling results with microbial community composition and associated metadata,
109 facilitating multi-dimensional analyses at scales that were previously difficult to achieve.

110 By lowering technical barriers, M2M-PostaViz broadens access to metabolic modelling
111 approaches and enables their application by non-specialists. To promote transparency,
112 reproducibility, and adoption, M2M-PostaViz is accompanied by an extensive tutorial and
113 curated test datasets, supporting users in applying metabolic modelling analyses to diverse
114 microbial ecosystems.

115 AI usage disclosure

116 AI tools (GitHub Copilot and GPT-5) were used occasionally for code generation, particularly
117 for debugging, and for grammar and language checks of the manuscript. All suggestions were
118 reviewed and validated by the human authors.

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